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Bayesian Deep Learning & Uncertainty estimation

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Bayesian Deep Learning – why?

- Ability to obtain uncertainty estimates
  - Improved decision making
  - Increased trust in ML systems
  - Responsible deployment of ML systems
  - Detecting unreliable inputs

Notation

- The machine learning model $g(\boldsymbol{x}; \boldsymbol{w})$ will be $p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{x})$ or at times $p_{\boldsymbol{w}}(\boldsymbol{y} \mid \boldsymbol{x})$.
- We will use $\mathcal{D} = \{(\boldsymbol{x}_1, \boldsymbol{y}_1), \ldots, (\boldsymbol{x}_N, \boldsymbol{y}_N)\}$ to denote the labeled training data. In other lectures, the notation $\boldsymbol{z} = (\boldsymbol{x}, y)$ for a pair and $\boldsymbol{Z} = (\boldsymbol{X}, \boldsymbol{y})$ for the full supervised dataset is used – we switch to $\mathcal{D}$ to avoid confusion with latent variables $\boldsymbol{z}$ or $\boldsymbol{Z}$
- The prediction $\hat{y}$ will become $\hat{p}$ or $\hat{p}_k$.
- A new data point that we aim to predict will be denoted as $(\boldsymbol{x}^{\text{test}}, \boldsymbol{y}^{\text{test}})$.
- A model will be denoted with $\mathcal{M}$

Math background

- Basic knowledge about Bayes Theorem and its use for parameter estimation for ML models; Bayesian model average
- Definitions of
  - Entropy H
  - Cross-Entropy CE
  - Mutual information I
  - KL divergence KL

Likelihood, Prior, Posterior, Evidence

- **Maximum likelihood:** find the most likely parameters given data

$$L(\mathcal{D}; \boldsymbol{w}) = p(\mathcal{D} \mid \boldsymbol{w})$$

- However, for complex models this leads to overfitting. Therefore, some $\boldsymbol{w}$ should be more probable than others
  - Prior distribution over parameters: $p(\boldsymbol{w})$
  - Bayes formula $p(\boldsymbol{w} \mid \mathcal{D}) = \dfrac{p(\mathcal{D} \mid \boldsymbol{w})\ p(\boldsymbol{w})}{p_{\boldsymbol{w}}(\mathcal{D})}$
  - Important note! Usually: $p(\mathcal{D}) \neq p_{\boldsymbol{w}}(\mathcal{D})$
- Maximum A Posteriori: (last term does not depend on params)

$$-\log p(\boldsymbol{w} \mid \mathcal{D}) = -\log p(\mathcal{D}|\boldsymbol{w}) - \log p(\boldsymbol{w}) + \log p_{\boldsymbol{w}}(\mathcal{D})$$

Maximum A Posteriori (MAP) approach for Deep Nets: a usual case

- Usually, the aim is to find the most likely parameters given data

$$\begin{aligned}\tilde{\boldsymbol{w}} &= \operatorname*{argmax}_{\boldsymbol{w}} p(\boldsymbol{w} \mid \boldsymbol{x}, \boldsymbol{y}) \\ &= \operatorname*{argmin}_{\boldsymbol{w}} -\log p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}) - \log p(\boldsymbol{w})\end{aligned}$$

- For example, softmax & cross-entropy at output and L2 regularization lead to

$$\tilde{\boldsymbol{w}} = \operatorname*{argmin}_{\boldsymbol{w}} \sum_k y_k \log(\hat{p}_k) + \lambda \|\boldsymbol{w}\|^2$$

- Note: $p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w})$ represents uncertainty about the label. However, we can make only a single prediction with the parameters we found

- Still a point estimate! No full Bayesian treatment! We want a full distribution over parameters!

A Bayesian approach: a distribution of parameters is wanted; supervised!

- Bayesian setting:

$$\boxed{p(\boldsymbol{w} \mid \boldsymbol{x}, \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{x}) \cdot p(\boldsymbol{w})}{\int_{\mathcal{W}} p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{x}) \cdot p(\boldsymbol{w}) d\boldsymbol{w}} = \frac{p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{x}) \cdot p(\boldsymbol{w})}{p(\boldsymbol{y} \mid \boldsymbol{x})}}$$

- Posterior: $p(\boldsymbol{w} \mid \boldsymbol{x}, \boldsymbol{y})$
- Prior: $p(\boldsymbol{w})$
- Likelihood: $p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{x})$
- Evidence: $\int_{\mathcal{W}} p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{x}) \cdot p(\boldsymbol{w}) d\boldsymbol{w}$
  - Usually intractable; especially for DNNs
  - If all other distributions are Gaussian: tractable. “Gaussian process”

Example: Gaussian data, Gaussian prior

- Data: $x \mid \mu \sim \mathcal{N}(\mu, \sigma_x^2)$ Pior: $\mu \sim \mathcal{N}(\nu, \sigma_\mu^2)$
- Parameter posterior

$$p(\mu \mid x) = \frac{p(x \mid \mu)p(\mu)}{\int_{-\infty}^{\infty} p(x \mid m)p(m)dm} \propto p(x \mid \mu)p(\mu)$$

$$p(x \mid \mu)p(\mu) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-\frac{(x-\mu)^2}{2\sigma_x^2}} \frac{1}{\sqrt{2\pi\sigma_\mu^2}} e^{-\frac{(\mu-\nu)^2}{2\sigma_\mu^2}}$$

- **Completing squares in exponent** and assuming normalized distributions, we see that this is again a Gaussian

$$\mu \mid x \sim \mathcal{N}\left(\frac{\mu\sigma_\mu^2 + \nu\sigma_x^2}{\sigma_\mu^2 + \sigma_x^2}, \frac{1}{1/\sigma_x^2 + 1/\sigma_\mu^2}\right)$$

Intuition?

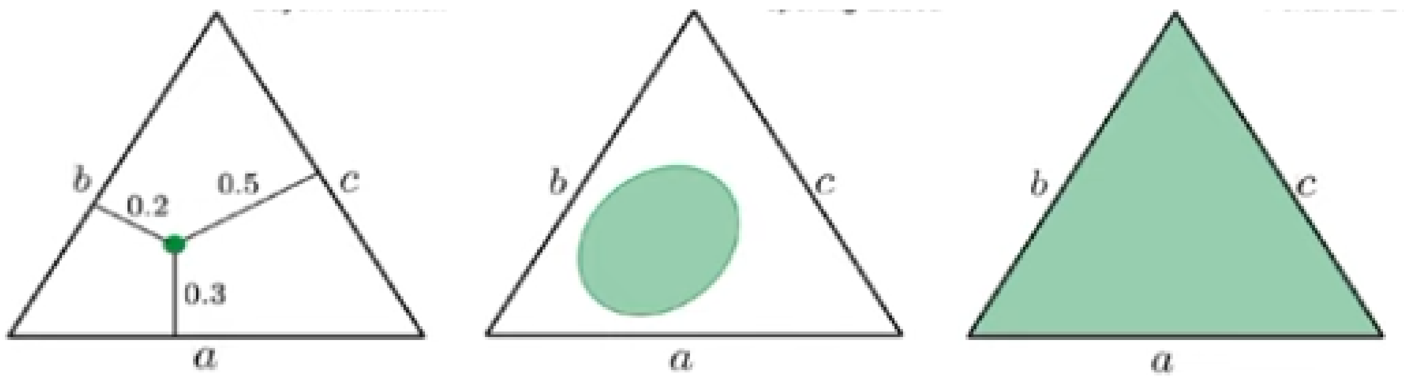
- Assume you train a vision model $p(y \mid \boldsymbol{x}, \boldsymbol{w})$ that should predict binary classes (“cat” vs “dog”).
- It provides for a particular image:

$$p(y = 1 \mid \boldsymbol{x}, \boldsymbol{w}) = 0.5$$

- Is the model uncertain about it’s prediction?
- Correct answer: we don’t know!
  - **A)** It could be the perfect model, but the image is ambiguous the prediction is correct: every time one labels the image it is labeled with 50% cat and 50% dog; $\rightarrow$ **aleatoric uncertainty**
  - **B)** The model is garbage and the image is indeed shows a cat (or: dog); uncertainty about the parameters $\rightarrow$ **epistemic uncertainty**
- **Does the model not know anything (B)? Or**
- **Does it know that it does not know anything (A).**

Types of uncertainy

- Levels of uncertainty representations



- Note: the softmax output of a neural network “pretends” to have no *epistemic uncertainty* and to just provide *aleatoric uncertainty*

Eyke Hüllermeier: "Uncertainty Quantification in Machine Learning: From Aleatoric to Epistemic I"; https://www.youtube.com/watch?v=vvUUmk9qfuA
Malinin, A., & Gales, M. (2018). Predictive uncertainty estimation via prior networks. Advances in neural information processing systems, 31.

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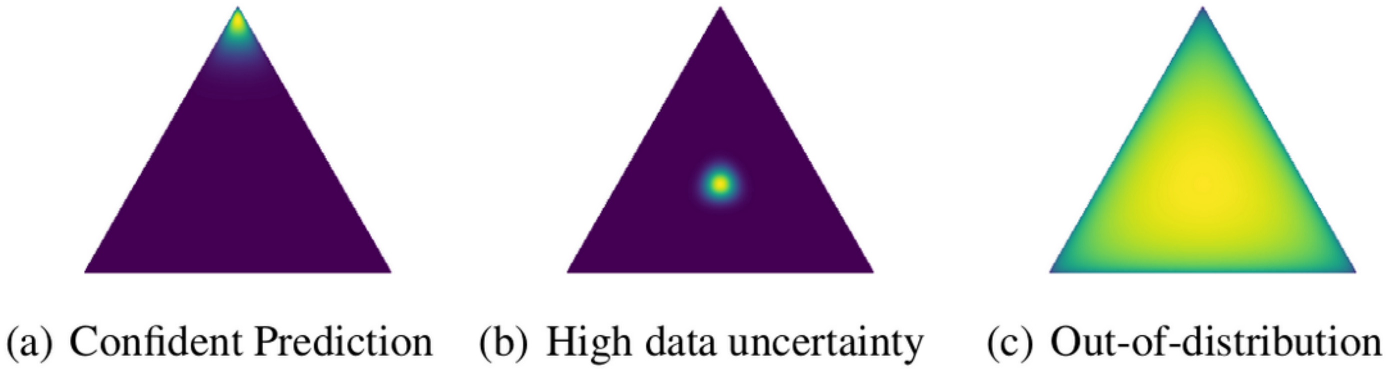

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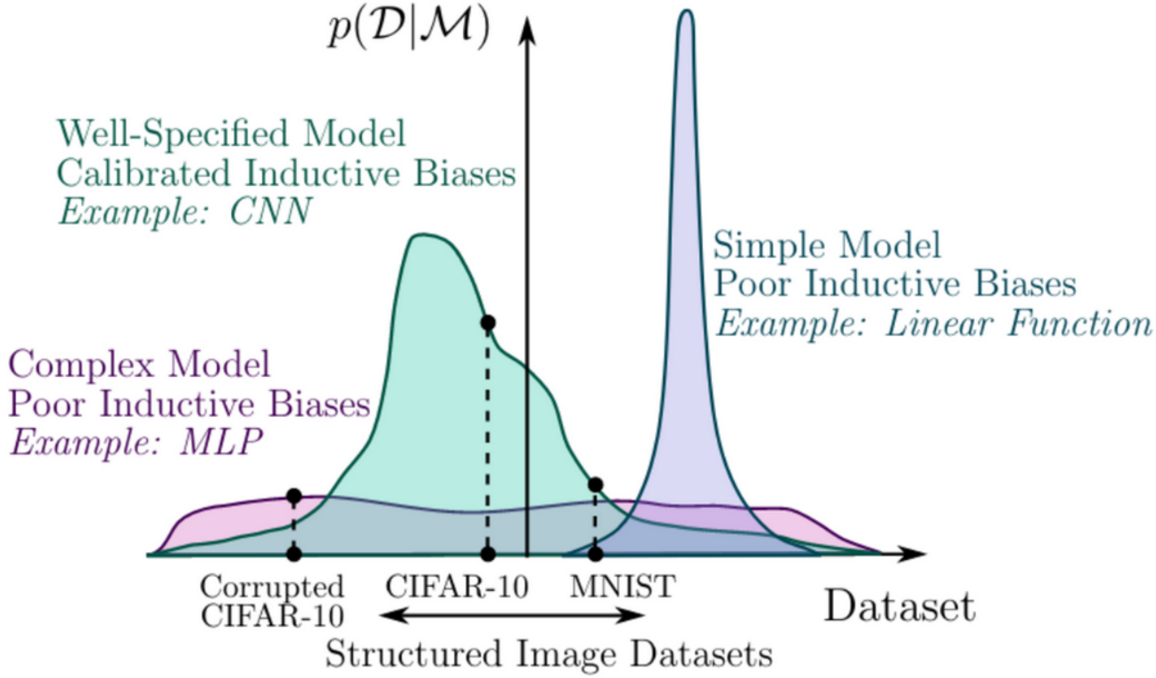
Sources of uncertainty

- Goal: have a **good model** for $p(\boldsymbol{y} \mid \boldsymbol{x})$; this would give us **aleatoric uncertainty**
  - Model $p(\boldsymbol{y} \mid \boldsymbol{x}; \boldsymbol{w})$
  - Model will never be perfect → **epistemic uncertainty** wanted
- Sources of epistemic uncertainty
  - A) No access to $p(\boldsymbol{x},\boldsymbol{y})$, thus to get $\boldsymbol{w}$, we need data
    - Drawing data from joint distribution $D \simeq p(\boldsymbol{x},\boldsymbol{y})$
    - Uncertainty in the drawn data (usually: no control over; IID)
  - B) Uncertainty about definition of hypothesis space
    - Usually hard to model
    - Expressive enough? Often has to be assumed
      - Assumption: $p(\boldsymbol{y} \mid \boldsymbol{x}) = p(\boldsymbol{y} \mid \boldsymbol{x}; \boldsymbol{w}^*)$; $\boldsymbol{w}^*$ is the parameter-vector of the true model
  - C) Uncertainty about chosen parameters $\boldsymbol{w}$
    - Bayesian framework: posterior distribution
    - Consider all possible $\boldsymbol{w}$, weighted by posterior $p(\boldsymbol{w} \mid D)$ –> intractable; Methods needed

B) Neural network generalization



- How is model class performance (~*inductive bias*) distributed over the range of all possible datasets (*support*)

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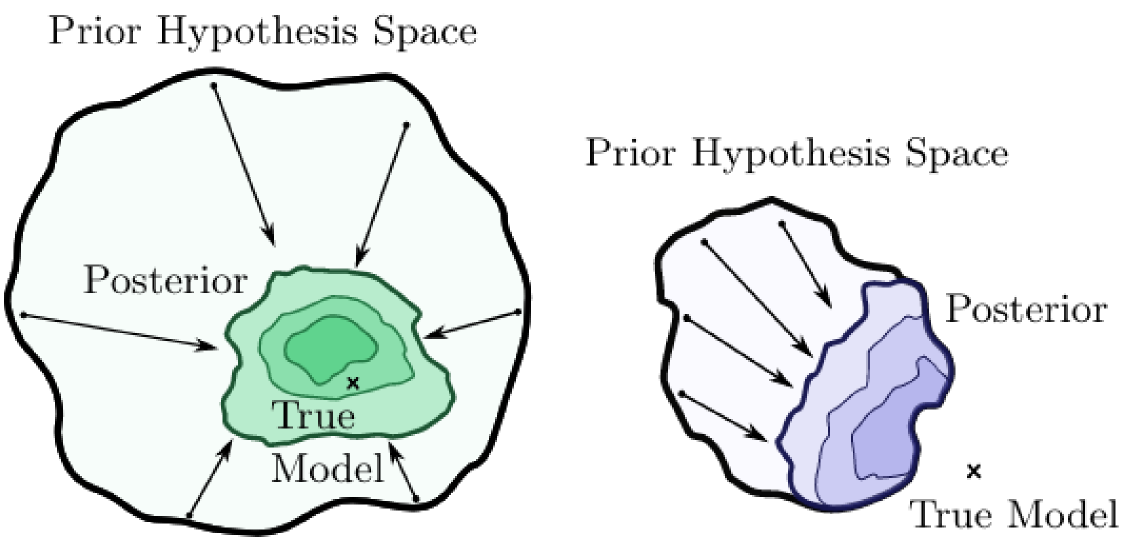
B)→ C) A probabilistic perspective of generalization



(b) (c) (d)

Key difference between frequentist and Bayesian approach: marginalization

- Posterior predictive distribution / distribution of outputs:

$$
\begin{aligned}
p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{x}^{\text{test}}, \mathcal{D}) &= \mathbb{E}_{\boldsymbol{w} \sim p(\boldsymbol{w} \mid \mathcal{D})} \left[p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{w}, \boldsymbol{x}^{\text{test}}) \right], \\
&= \int_{\mathcal{W}} p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{w}, \boldsymbol{x}^{\text{test}}) p(\boldsymbol{w} \mid \mathcal{D}) d\boldsymbol{w}
\end{aligned}
$$

- Defines probability for class label given input and dataset
- Marginalization over parameters
- “Bayesian model averaging”

Inference methods

- Maximum likelihood

$$\boldsymbol{w}_{\mathrm{ML}} = \mathrm{argmax}_{\boldsymbol{w}}\ p(\mathcal{D} \mid \boldsymbol{w})$$

- Maximum A-posteriori (MAP) inference

$$\boldsymbol{w}_{\mathrm{MAP}} = \mathrm{argmax}_{\boldsymbol{w}}\ p(\boldsymbol{w} \mid \mathcal{D}) = \mathrm{argmax}_{\boldsymbol{w}}\ p(\mathcal{D} \mid \boldsymbol{w})(\boldsymbol{w})$$

- Bayesian and approximate Bayesian inference

$$\int_{\boldsymbol{w}} F(\boldsymbol{w}) p(\boldsymbol{w} \mid \mathcal{D}) d\boldsymbol{w} \approx \frac{1}{M} \sum_{m=1}^{M} F(\boldsymbol{w}_k); \quad \boldsymbol{w}_k \sim p(\boldsymbol{w} \mid \mathcal{D})$$

Definition of uncertainty

- *Total uncertainty* often defined as entropy of the predictive distribution (e.g. Gal, 2016; Hüllermeier, 2021):

$$\mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \mathcal{D})\right] \ = \ \int_{\mathcal{W}} \mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})\right] \ p(\tilde{\boldsymbol{w}} \mid \mathcal{D}) \ \mathrm{d}\tilde{\boldsymbol{w}} \ + \ \mathrm{I}(Y, W) \ .$$

- The *aleatoric uncertainty* is given by

$$\mathbb{E}_{p(\tilde{\boldsymbol{w}} \mid \mathcal{D})}\left[\mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})\right]\right] \ = \ \int_{\mathcal{W}} \mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})\right] \ p(\tilde{\boldsymbol{w}} \mid \mathcal{D}) \ \mathrm{d}\tilde{\boldsymbol{w}}$$

- *Epistemic uncertainty* obtained as difference

$$\begin{aligned} \mathrm{I}(Y;W) \ &= \ \mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \mathcal{D})\right] \ - \ \mathbb{E}_{p(\tilde{\boldsymbol{w}} \mid \mathcal{D})}\left[\mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})\right]\right] \\ &= \ \int_{\mathcal{W}} \left(\mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \mathcal{D})\right] \ - \ \mathrm{H}\left[p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})\right]\right) \ p(\tilde{\boldsymbol{w}} \mid \mathcal{D}) \ \mathrm{d}\tilde{\boldsymbol{w}} \end{aligned}$$

$$p(\boldsymbol{y} \mid \boldsymbol{x}, \mathcal{D}) \ = \ p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}^*) \ = \int_{\mathcal{W}} p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}}) \ p(\tilde{\boldsymbol{w}} \mid \mathcal{D}) \ \mathrm{d}\tilde{\boldsymbol{w}}$$

Intuition?

- Assume you train a vision model $p(y \mid \boldsymbol{x}, \boldsymbol{w})$ that should predict binary classes (“cat” vs “dog”).
- It provides for a particular image:

$$p(y = 1 \mid \boldsymbol{x}, \boldsymbol{w}) = 0.5$$

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- Correct answer: we don’t know!
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Monte-Carlo Approximation

- How to approximate the integral? → sampling from posterior

$$p(\boldsymbol{y} \mid \boldsymbol{x}, \mathcal{D}) \;=\; p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}^*) \;= \int_{\mathcal{W}} p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})\; p(\tilde{\boldsymbol{w}} \mid \mathcal{D})\; \mathrm{d}\tilde{\boldsymbol{w}} \approx \frac{1}{K} \sum_{k=1}^{K} p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}_k); \quad \boldsymbol{w}_k \sim p(\tilde{\boldsymbol{w}} \mid \mathcal{D})$$

- How to sample from posterior?
  - METHODS (see later)

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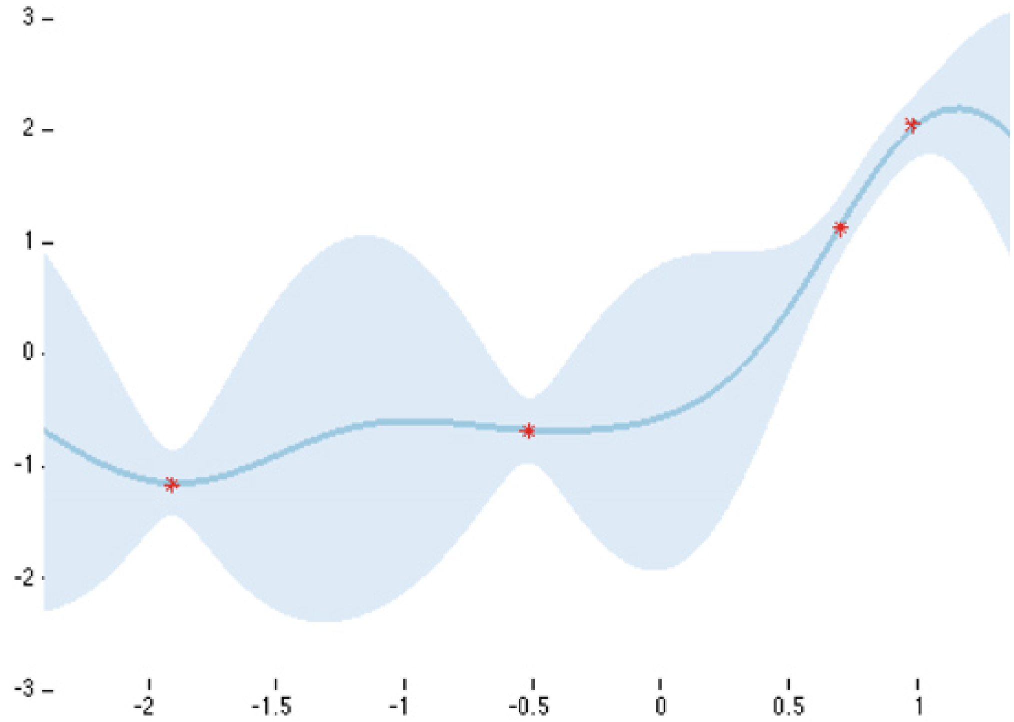
Full predictive distribution

- Linear regression example
- Uncertainty close to data points is small
- High uncertainty elsewhere



A selection of Bayesian DL approaches

- 15.1. Deep Ensembles
- 15.2. Weight uncertainty in DNNs (variational approach)
- 15.3. Monte-Carlo Dropout
- 15.4. Hessian-based approach

15.1. Deep Ensembles

- Main idea: approximate expectation with averages

$$p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{x}^{\text{test}}, \mathcal{D}) = \mathbb{E}_{\boldsymbol{w} \sim p(\boldsymbol{w} \mid \mathcal{D})} \left[p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{w}, \boldsymbol{x}^{\text{test}}) \right]$$

- Train several DNNs with different random seeds, obtain parameters $\boldsymbol{w}^{(m)}$ and take average

$$p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{x}^{\text{test}}, \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^{M} p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{w}^{(m)}, \boldsymbol{x}^{\text{test}},)$$

- Averages can be weighted differently
- Appears as simple technique, but performs well in practice

Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017, December). Simple and scalable predictive uncertainty estimation using deep ensembles. In Proceedings of the 31st International Conference on Neural Information Processing Systems (pp. 6405-6416).

Deep Ensembles

Algorithm 1 Pseudocode of the training procedure for our method

$\triangleright$ *Let each neural network parametrize a distribution over the outputs, i.e.* $p_\theta(y|\mathbf{x})$. *Use a proper scoring rule as the training criterion* $\ell(\theta, \mathbf{x}, y)$. *Recommended default values are* $M = 5$ *and* $\boldsymbol{\epsilon} = 1\%$ *of the input range of the corresponding dimension (e.g 2.55 if input range is [0,255]).*
Initialize $\theta_1, \theta_2, \ldots, \theta_M$ randomly
for $m = 1 : M$ **do** $\triangleright$ *train networks independently in parallel*
  Sample data point n_m randomly for each net $\triangleright$ *single* n_m *for clarity, minibatch in practice*
  Generate adversarial example using $\mathbf{x}'_{n_m} = \mathbf{x}_{n_m} + \boldsymbol{\epsilon}\, \mathrm{sign}\big(\nabla_{\mathbf{x}_{n_m}} \ell(\theta_m, \mathbf{x}_{n_m}, y_{n_m})\big)$
  Minimize $\ell(\theta_m, \mathbf{x}_{n_m}, y_{n_m}) + \ell(\theta_m, \mathbf{x}'_{n_m}, y_{n_m})$ w.r.t. θ_m $\triangleright$ *adversarial training (optional)*

Notation in code
does not
fit to our notation!

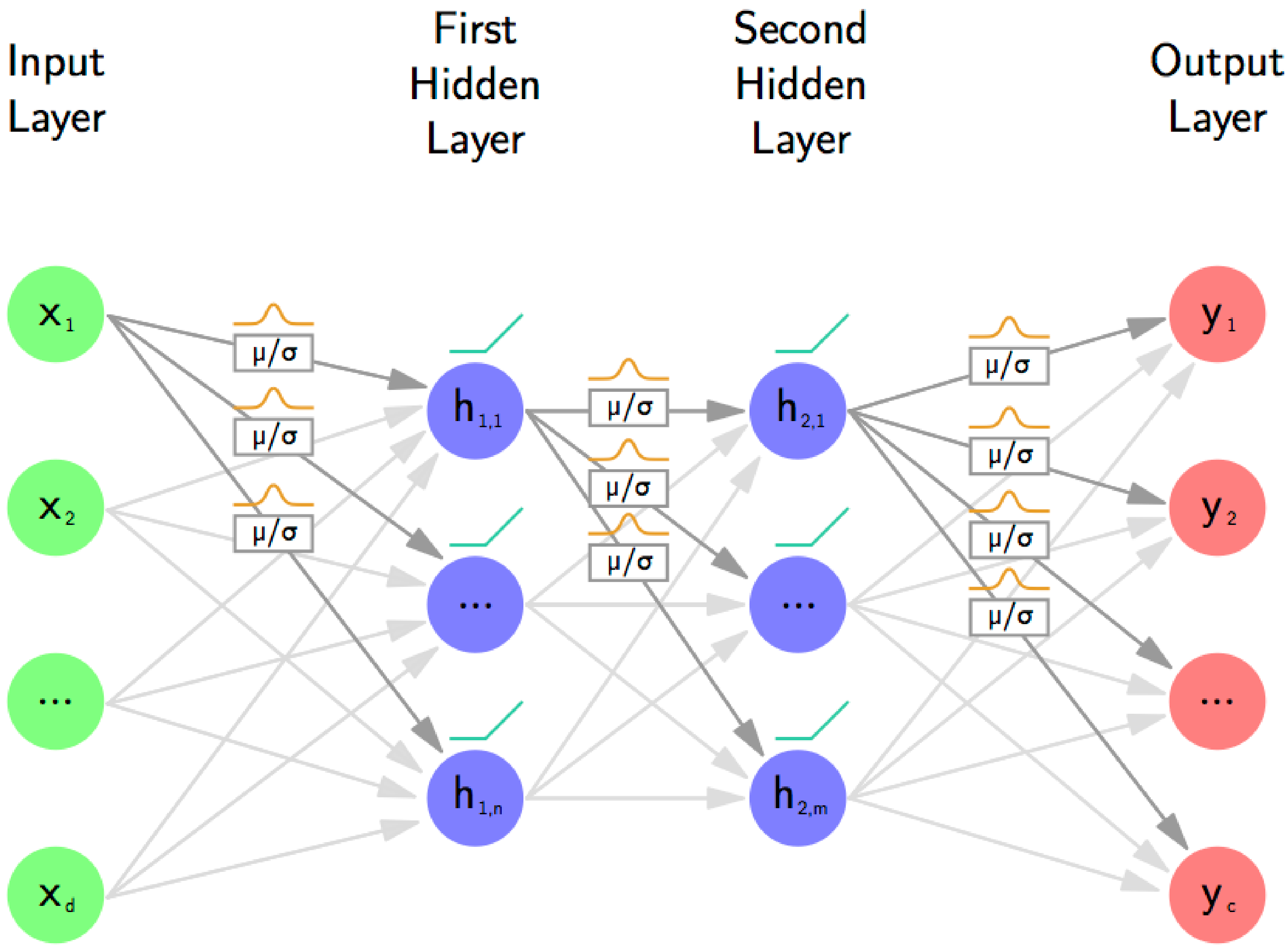
Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017, December). Simple and scalable predictive uncertainty estimation using deep ensembles. In Proceedings of the 31st International Conference on Neural Information Processing Systems (pp. 6405-6416).

15.2. Variational approach: Weight Uncertainty in DNNs

- Approximate predictive distribution
- Approach: Weight uncertainty in neural networks (Blundell et al., 2015).
- Variational approach, learning finds the parameters $\boldsymbol{w}$ of a parameterized distribution of the weights $q(\boldsymbol{w} \mid \boldsymbol{\theta})$
- New new set of parameters $\boldsymbol{\theta}$ that we did not have before.
  - parameters determine the distribution of weights,
  - e.g. mean and standard deviation of Gaussian

Blundell, C., Cornebise, J., Kavukcuoglu, K., & Wierstra, D. (2015, June). Weight uncertainty in neural network. In International Conference on Machine Learning (pp. 1613-1622). PMLR.

Weight Uncertainty in DNNs
Intuition



Weight Uncertainty in DNNs

- Gaussian approximation of the parameter posterior
- Minimization of KL divergence with true posterior

$$\begin{aligned}
\tilde{\boldsymbol{\theta}} &= \operatorname{argmin}_{\boldsymbol{\theta}} \boxed{\mathrm{KL}\left(q(\boldsymbol{w} \mid \boldsymbol{\theta}) \;||\; p(\boldsymbol{w} \mid \mathcal{D})\right)} \\
&= \operatorname{argmin}_{\boldsymbol{\theta}} \int_{\mathcal{W}} q(\boldsymbol{w} \mid \boldsymbol{\theta}) \log \frac{q(\boldsymbol{w} \mid \boldsymbol{\theta})}{p(\boldsymbol{w}) p(\mathcal{D} \mid \boldsymbol{w})} d\boldsymbol{w} \\
&= \operatorname{argmin}_{\boldsymbol{\theta}} \boxed{\mathrm{KL}\left(q(\boldsymbol{w} \mid \boldsymbol{\theta}) \;||\; p(\boldsymbol{w})\right)} - \mathbb{E}_{\boldsymbol{w} \sim q(\boldsymbol{w} \mid \boldsymbol{\theta})} [\log p(\mathcal{D} \mid \boldsymbol{w})]
\end{aligned}$$

Weight Uncertainty in DNNs

- Objective function (ELBO):

$$L(\boldsymbol{\theta}, \mathcal{D}) = L(\boldsymbol{\theta}, \boldsymbol{X}, \boldsymbol{Y}) = \mathrm{KL}\left(q(\boldsymbol{w} \mid \boldsymbol{\theta}) \mid\mid p(\boldsymbol{w})\right) - \mathbb{E}_{\boldsymbol{w} \sim q(\boldsymbol{w} \mid \boldsymbol{\theta})}[\log p(\mathcal{D} \mid \boldsymbol{w})]$$

$$= \mathbb{E}_{\boldsymbol{w} \sim q(\boldsymbol{w} \mid \boldsymbol{\theta})}\left[\log q(\boldsymbol{w} \mid \boldsymbol{\theta}) - \log p(\boldsymbol{w})\right] - \mathbb{E}_{\boldsymbol{w} \sim q(\boldsymbol{w} \mid \boldsymbol{\theta})}[\log p(\mathcal{D} \mid \boldsymbol{w})]$$

  - Similar to objective in VAEs
  - Involves sampling from posterior estimates
  - How to approach this with gradient descent? We cannot backpropagate to a sampling procedure

Weight Uncertainty in DNNs: Reparametrization trick

- We assume that weights are determined by some function that has both a deterministic and a random part:

$$\boldsymbol{w} = h(\boldsymbol{\theta}, \boldsymbol{\epsilon})$$

  - ϵ: noise
  - $\boldsymbol{\theta}$: parameters of probability distribution
  - Similar to reparametrization trick in VAEs
  - E.g. mean and variance of Gaussian plus standard noise

Weight Uncertainty in DNNs: Reparametrization trick

- Backpropagation to parameters of probability distribution

$$
\begin{aligned}
\frac{\partial}{\partial \boldsymbol{\theta}} \mathbb{E}_{\boldsymbol{w} \sim q(\boldsymbol{w}|\theta)}[f(\boldsymbol{w}, \boldsymbol{\theta})] &= \frac{\partial}{\partial \boldsymbol{\theta}} \int f(\boldsymbol{w}, \boldsymbol{\theta}) q(\boldsymbol{w} \mid \boldsymbol{\theta}) d\boldsymbol{w} = \\
&= \frac{\partial}{\partial \boldsymbol{\theta}} \int f(\boldsymbol{w}, \boldsymbol{\theta}) q(\boldsymbol{\epsilon}) d\boldsymbol{\epsilon} \\
&= \mathbb{E}_{\boldsymbol{\epsilon} \sim q(\boldsymbol{\epsilon})} \left[\frac{\partial f(\boldsymbol{w}, \boldsymbol{\theta})}{\partial \boldsymbol{w}} \frac{\partial \boldsymbol{w}}{\partial \boldsymbol{\theta}} + \frac{\partial f(\boldsymbol{w}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right]
\end{aligned}
$$

- We used Leibnitz integral rule to switch integration and differentiation
- We use: $f(\boldsymbol{w}, \boldsymbol{\theta}) = \log q(\boldsymbol{w} \mid \boldsymbol{\theta}) - \log p(\boldsymbol{w}) - \log p(\mathcal{D} \mid \boldsymbol{w})$

Weight Uncertainty in DNNs: Reparametrization trick

- With $f(\boldsymbol{w}, \boldsymbol{\theta}) = \log q(\boldsymbol{w} \mid \boldsymbol{\theta}) - \log p(\boldsymbol{w}) - \log p(\mathcal{D} \mid \boldsymbol{w})$
  we approximate the objective function by

$$L(\boldsymbol{\theta}, \mathcal{D}) \approx \sum_{m=1}^{M} \log q(\boldsymbol{w}^{(m)} \mid \boldsymbol{\theta}) - \log p(\boldsymbol{w}^{(m)}) - \underbrace{\log p(\mathcal{D} \mid \boldsymbol{w}^{(m)})}_{\text{usual obj of a DNN}}$$

- Where $\boldsymbol{w}^{(m)}$ is a parameter vector randomly drawn from the estimated posterior distribution $q(\boldsymbol{w} \mid \boldsymbol{\theta})$

Weight Uncertainty in DNNs: Algorithm

- Sample $\varepsilon \sim \mathcal{N}(0,1)$,
- set $\boldsymbol{w} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$
- let $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\sigma})$,
- let $f(\boldsymbol{w}, \boldsymbol{\theta}) = \log q(\boldsymbol{w} \mid \boldsymbol{\theta}) - \log p(\boldsymbol{w}) - \log p(\mathcal{D} \mid \boldsymbol{w})$,
- calculate gradients of $f(\boldsymbol{w}, \boldsymbol{\theta})$ with respect to $\boldsymbol{\mu}$ using backprop:

$$\Delta_{\boldsymbol{\mu}} = \frac{\partial f(\boldsymbol{w}, \boldsymbol{\theta})}{\partial \boldsymbol{w}} \odot \mathbf{1} + \frac{\partial f(\boldsymbol{w}, \boldsymbol{\theta})}{\partial \boldsymbol{\mu}}$$

- calculate gradients of $f(\boldsymbol{w}, \boldsymbol{\theta})$ with respect to $\boldsymbol{\sigma}$ using backprop:

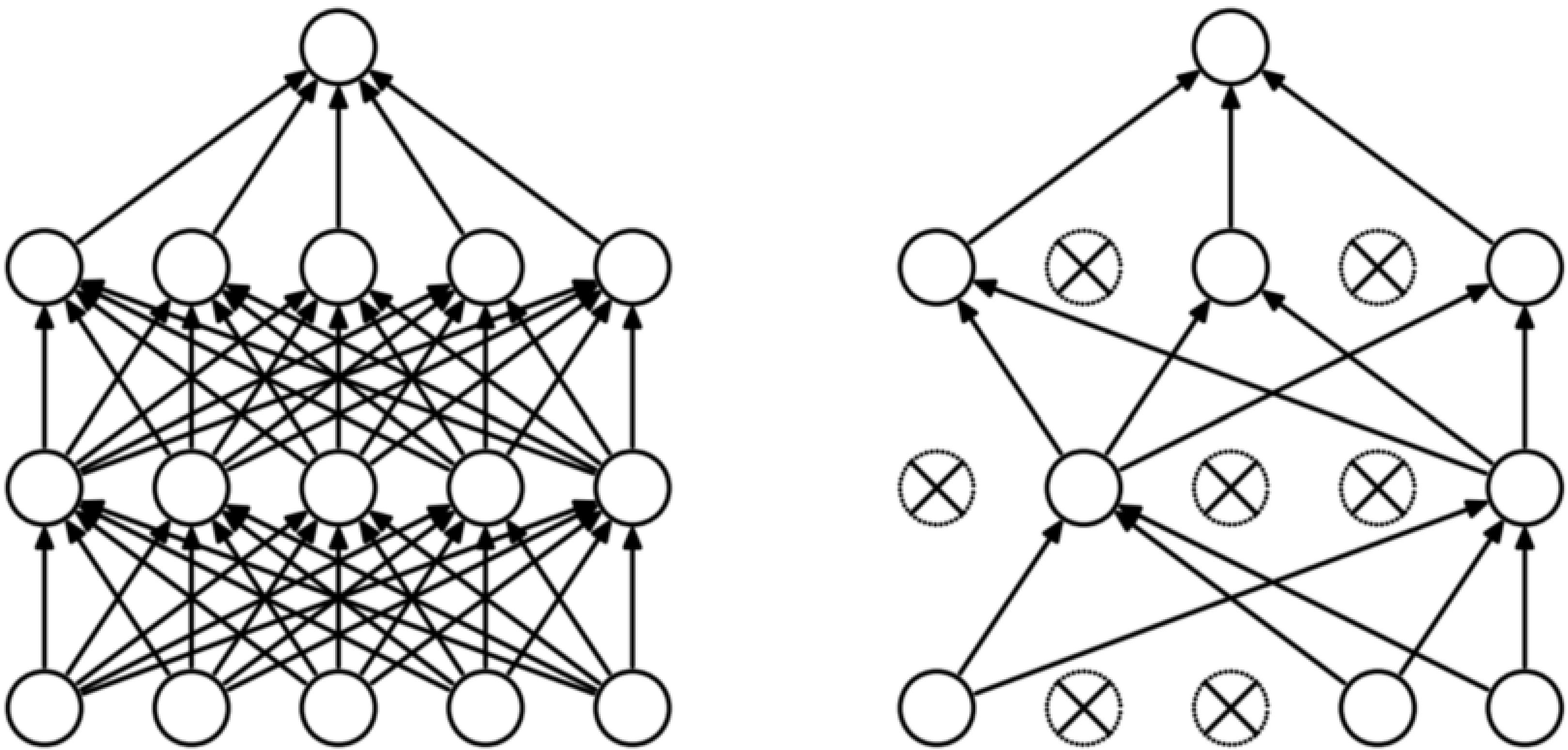
$$\Delta_{\boldsymbol{\sigma}} = \frac{\partial f(\boldsymbol{w}, \boldsymbol{\theta})}{\partial \boldsymbol{w}} \odot \boldsymbol{\varepsilon} + \frac{\partial f(\boldsymbol{w}, \boldsymbol{\theta})}{\partial \boldsymbol{\sigma}}$$

- and update the variational parameters:

$$\boldsymbol{\mu}^{\text{new}} = \boldsymbol{\mu}^{\text{old}} - \eta \Delta_{\boldsymbol{\mu}}$$
$$\boldsymbol{\sigma}^{\text{new}} = \boldsymbol{\sigma}^{\text{old}} - \eta \Delta_{\boldsymbol{\sigma}}$$

These are
the usual gradients
from backprop

15.3. Monte Carlo Dropout



(a) Standard Neural Net (b) After applying dropout.

Monte-Carlo Dropout

- Similar derivation to *Weight Uncertainty in DNNs*
- Use of the equivalence to Gaussian processes
- Consider a two-layer network
- Intractable posterior; approximation with $q(\boldsymbol{w})$
  - Weights assumed to come from mixture

$$q(\mathbf{W}_1) = \prod_{q=1}^{Q} q(\mathbf{w}_q),$$

$$q(\mathbf{w}_q) = p_1 \mathcal{N}(\mathbf{m}_q, \boldsymbol{\sigma}^2 \mathbf{I}_K) + (1 - p_1)\mathcal{N}(0, \boldsymbol{\sigma}^2 \mathbf{I}_K)$$

  - Applying Dropout is like drawing from this posterior estimates
  - Objective is again ELBO approximated with sampling

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MC Dropout

- Two-layer network (one hidden layer):

$$p(y \mid \boldsymbol{x}, \boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}) = \mathrm{softmax}(\boldsymbol{W}^{[2]} f(\boldsymbol{W}^{[1]}\boldsymbol{x}))$$

- Gaussian prior on weights

$$p(W^{[1]}) = \mathcal{N}(0, \boldsymbol{I}) \quad \text{and} \quad p(W^{[2]}) = \mathcal{N}(0, \boldsymbol{I})$$

- Full predictive distribution

$$p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{x}^{\text{test}}, \mathcal{D}) = \int_{\mathcal{W}} p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}, \boldsymbol{x}^{\text{test}}) \; \underbrace{p(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]} \mid \mathcal{D})}_{\text{untractable}} d\boldsymbol{W}^{[1]} d\boldsymbol{W}^{[2]}$$

- Minimizing KL-divergence and connection to ELBO (see VAEs)

$$\boxed{D_{KL}(q(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}) || p(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]} \mid \mathcal{D}))}$$

$$\boxed{D_{KL}(q(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}) || p(\boldsymbol{W}^{[1]}) p(\boldsymbol{W}^{[2]})) - \sum_{n=1}^{N} \int q(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}) \log(y_n \mid x_n, \boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]})}$$

- With variational distribution; mixture of Gaussians

$$q(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}) = q(\boldsymbol{W}^{[1]}) q(\boldsymbol{W}^{[2]})$$

$$q(\boldsymbol{W}) = p\mathcal{N}(\boldsymbol{M}, \sigma^2 \boldsymbol{I}) + (1-p)\mathcal{N}(0, \sigma^2 \boldsymbol{I})$$

MC Dropout

- KL divergence between variational and prior

$$D_{KL}(q(\boldsymbol{W}^{[l]})||p(\boldsymbol{W}^{[l]})) \approx dI(\sigma^2 - \log(\sigma^2) - 1) + p/2||\boldsymbol{W}^{[l]}||_2^2 + \mathrm{const}$$

- Hence: weight matrices sampled from

$$\boldsymbol{W}^{[l]} \approx \mathrm{diag}(\boldsymbol{z}^{[l]})\boldsymbol{M}^{[l]}, \boldsymbol{z} \sim \mathrm{Bernoulli}(p)$$

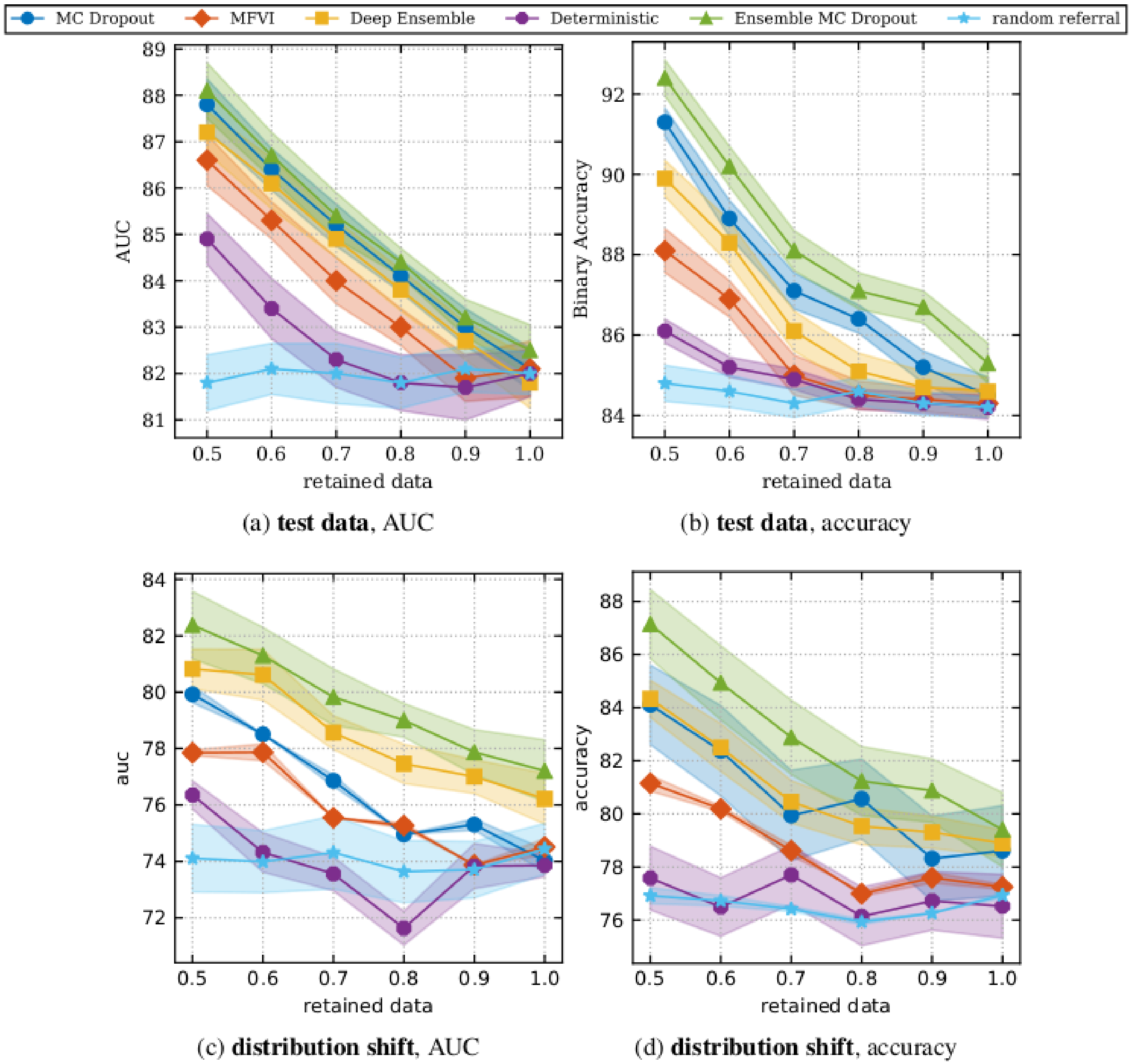
- Objective:

$$D_{KL}(q(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}||p(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]} \mid \mathcal{D})) \leq -\sum_{n=1}^{N} \log p(y_n \mid x_n, \boldsymbol{W}_n^{[1]}, \boldsymbol{W}_n^{[2]}) + p^{[1]}/2||\boldsymbol{M}_1||_2^2 + p^{[2]}/2||\boldsymbol{M}_2||_2^2$$

where $\boldsymbol{W}_n = \mathrm{diag}(\boldsymbol{z}_n)\boldsymbol{M}, \quad \boldsymbol{z} \sim \mathrm{Bernoulli}(p)$ are Monte Carlo samples

Same loss as used when training with dropout! (and reg.)

Bayesian DL methods compared



(a) **test data**, AUC

(b) **test data**, accuracy

(c) **distribution shift**, AUC

(d) **distribution shift**, accuracy

Filos, A., Farquhar, S., Gomez, A. N., Rudner, T. G., Kenton, Z., Smith, L., ... & Gal, Y. (2019). A systematic comparison of bayesian deep learning robustness in diabetic retinopathy tasks. arXiv preprint arXiv:1912.10481.

15.4. Hessian-based approach

- Regression task: Gaussian noise on labels

$$p(y \mid \boldsymbol{x}, \boldsymbol{w}, \sigma_\epsilon^2) = \mathcal{N}(y \mid g(\boldsymbol{x}; \boldsymbol{w}), \sigma_\epsilon^2)$$

- Gaussian prior on weights

$$p(\boldsymbol{w} \mid \sigma_w^2) = \mathcal{N}(\boldsymbol{w} \mid \mathbf{0}, \sigma_w^2)$$

- Dataset $\mathcal{D}$ of data points and labels; posterior and log post.:

$$p(\boldsymbol{w} \mid \mathcal{D}, \sigma_w^2, \sigma_\epsilon^2) \propto p(\boldsymbol{w} \mid \sigma_w^2) p(\mathcal{D} \mid \boldsymbol{w}, \sigma_\epsilon^2)$$

$$\log p(\boldsymbol{w} \mid \mathcal{D}, \sigma_w^2, \sigma_\epsilon^2) = \frac{1}{2\sigma_w^2} \boldsymbol{w}^T \boldsymbol{w} - \frac{1}{2\sigma_\epsilon^2} \sum_{n=1}^{N} (g(\boldsymbol{x}; \boldsymbol{w}) - y_n)^2 + C$$

The last function is optimized with the usual gradient descent techniques and we obtain parameters: $\boldsymbol{w}_{\mathrm{MAP}}$

15.4. Hessian-based approach

- Having found $\boldsymbol{w}_{\mathrm{MAP}}$, we can make a local Gaussian approximation by evaluation the matrix of second derivatives of the negative log posterior:

$$\boldsymbol{A} = -\nabla^2 \log p(\boldsymbol{w} \mid \mathcal{D}, \sigma_w^2, \sigma_\epsilon^2) = 1/\sigma_w^2 \boldsymbol{I} + 1/\sigma_\epsilon^2 \boldsymbol{H}$$

where $\boldsymbol{H}$ is the usual Hessian of the loss function of the neural network. Can be approximated in various ways for small networks.

- Gaussian approximation of posterior

$$q(\boldsymbol{w} \mid \mathcal{D}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{w}_{\mathrm{MAP}}, \boldsymbol{A}^{-1})$$

- Predictive distribution (still intractable, sampling from approx)

$$p(y \mid \boldsymbol{x}, \mathcal{D}) = \int \mathcal{N}(y \mid g(\boldsymbol{x}, \boldsymbol{w}), \sigma_\epsilon^2) q(\boldsymbol{w} \mid \mathcal{D}) d\boldsymbol{w}$$

Intractable because distribution on weights is inside the neural net

15.5. Hessian-based approach

- If we would have a linear model instead of $g(\boldsymbol{x};\boldsymbol{w})$, this integral would be analytically tractable.
- Let us approximate the neural net around $\boldsymbol{w}_{\mathrm{MAP}}$ with a Taylor approximation up to first degree

$$g(\boldsymbol{x};\boldsymbol{w}) \approx g(\boldsymbol{x};\boldsymbol{w}_{\mathrm{MAP}}) + \boldsymbol{g}^T(\boldsymbol{w} - \boldsymbol{w}_{\mathrm{MAP}})$$

  where $\boldsymbol{g} = \nabla_{\boldsymbol{w}} g(\boldsymbol{x};\boldsymbol{w})|_{\boldsymbol{w}_{\mathrm{MAP}}}$

- Linear Gaussian model with Gaussian distribution of weights and prior whose mean is linear function of weights

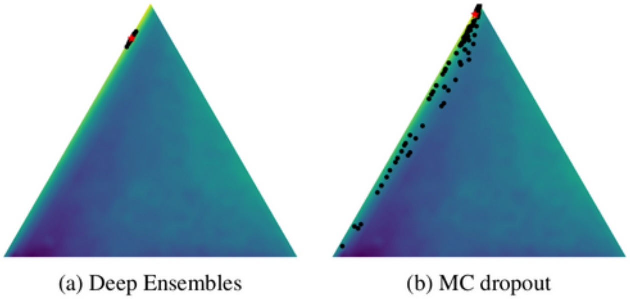
$$p(y \mid \boldsymbol{x}, \boldsymbol{w}, \sigma_\epsilon^2) \approx \mathcal{N}(y \mid g(\boldsymbol{x}, \boldsymbol{w}_{\mathrm{MAP}}) + \boldsymbol{g}^T(\boldsymbol{w} - \boldsymbol{w}_{\mathrm{MAP}}), \sigma_\epsilon^2)$$

  General results on marginal Gaussian provide:

$$p(y \mid \boldsymbol{x}, \mathcal{D}, \sigma_\epsilon^2, \sigma_w^2) \approx \mathcal{N}(y \mid g(\boldsymbol{x}, \boldsymbol{w}_{\mathrm{MAP}}), \sigma_\epsilon^2 + \boldsymbol{g}^T \boldsymbol{A} \boldsymbol{g})$$

Methods developed at JKU: Quantification of Uncertainty with Adversarial Models

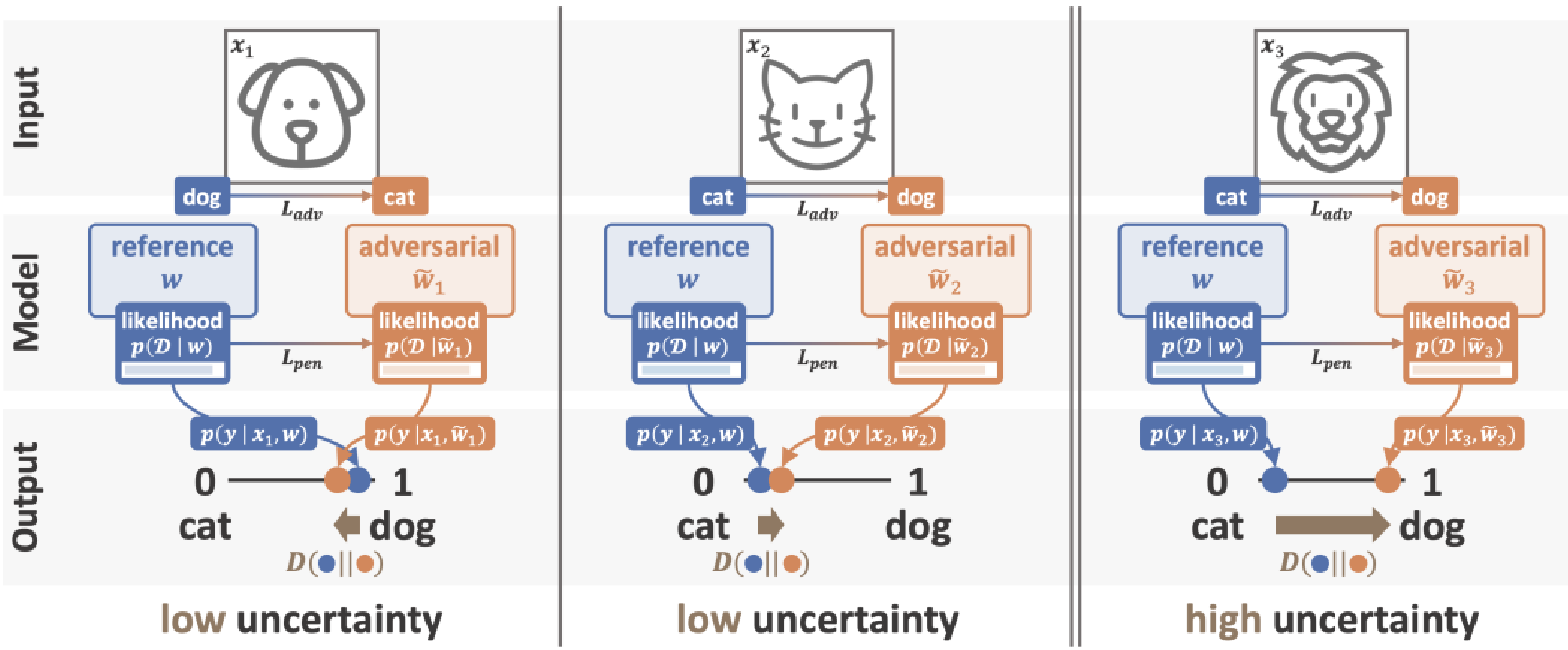
- **Problem:** how to find diverse models
  - Methods usually find similar models (through sgd)



- **Idea:** explicitly search for alternative models that also explain the data well (high posterior)
- Searching for modes in posterior
  - **Adversarial attacks** in weight space

Schweighofer, K., Aichberger, L., Ielanskyi, M., Klambauer, G., & Hochreiter, S. (2023). Quantification of uncertainty with adversarial models. Advances in Neural Information Processing Systems, 36, 19446-19484.

Methods developed at JKU:
Quantification of Uncertainty with Adversarial Models



Schweighofer, K., Aichberger, L., Ielanskyi, M., Klambauer, G., & Hochreiter, S. (2023). Quantification of uncertainty with adversarial models. Advances in Neural Information Processing Systems, 36, 19446-19484.

Methods developed at JKU: Quantification of Uncertainty with Adversarial Models

- Definition of *adversarial model*:

Definition 1. *Given are a new test data point* $\boldsymbol{x}$, *a reference conditional probability model* $p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w})$ *from a model class parameterized by* $\boldsymbol{w}$, *a divergence or distance measure* $\mathrm{D}(\cdot, \cdot)$ *for probability distributions,* $\gamma > 0$, $\Lambda > 0$, *and a dataset* $\mathcal{D}$. *Then a model with parameters* $\tilde{\boldsymbol{w}}$ *that satisfies the inequalities* $|\log p(\boldsymbol{w} \mid \mathcal{D}) - \log p(\tilde{\boldsymbol{w}} \mid \mathcal{D})| \leqslant \gamma$ *and* $\mathrm{D}(p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}), p(\boldsymbol{y} \mid \boldsymbol{x}, \tilde{\boldsymbol{w}})) \geq \Lambda$ *is called an* $(\gamma, \Lambda)-$*adversarial model.*

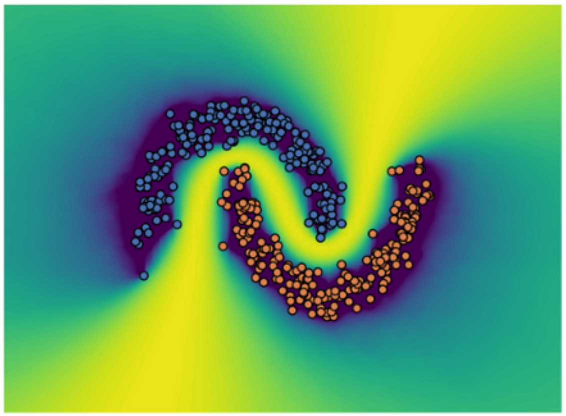
- Search for adversarial model:

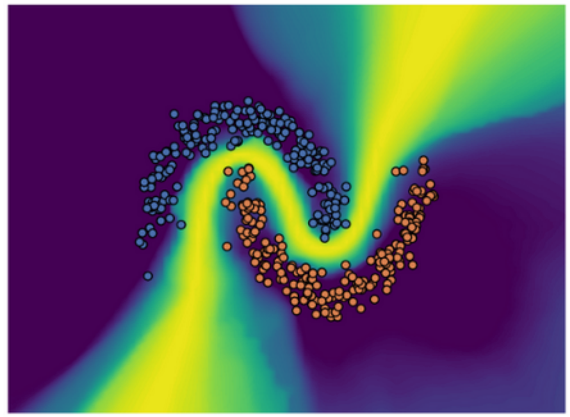
$$\max_{\boldsymbol{\delta} \in \Delta} \mathrm{D}(p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w}), p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{w} + \boldsymbol{\delta})) \quad \text{s.t. } \log p(\boldsymbol{w} \mid \mathcal{D}) - \log p(\boldsymbol{w} + \boldsymbol{\delta} \mid \mathcal{D}) \leqslant \gamma .$$

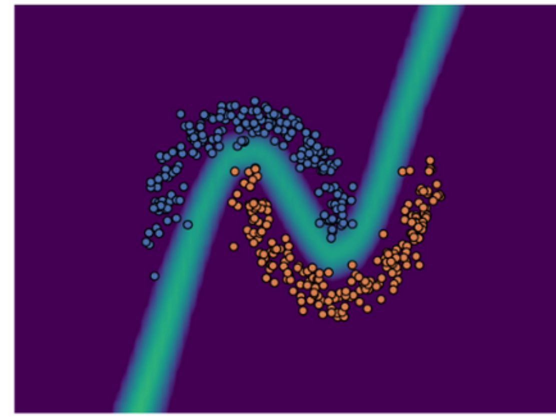
- Iterative procedure (gradient-based) employed to search for these models

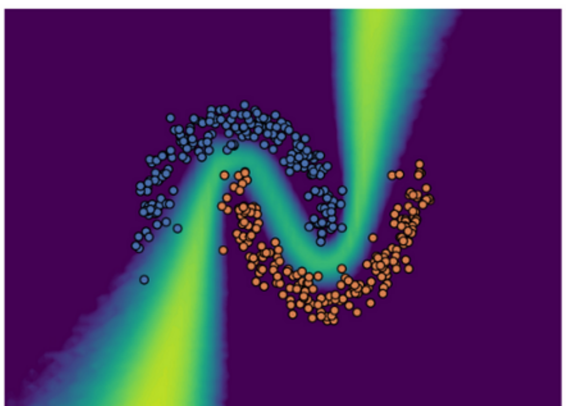
Schweighofer, K., Aichberger, L., Ielanskyi, M., Klambauer, G., & Hochreiter, S. (2023). Quantification of uncertainty with adversarial models. Advances in Neural Information Processing Systems, 36, 19446-19484.

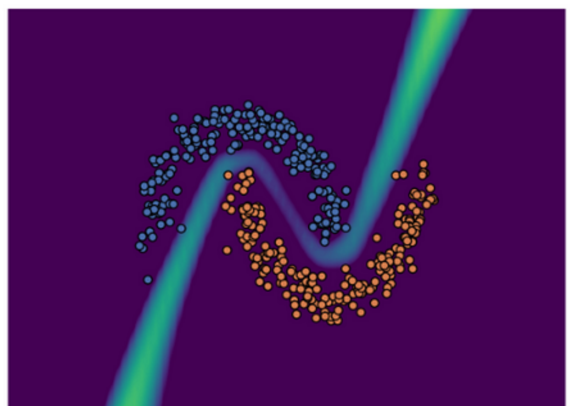
Methods developed at JKU: Quantification of Uncertainty with Adversarial Models

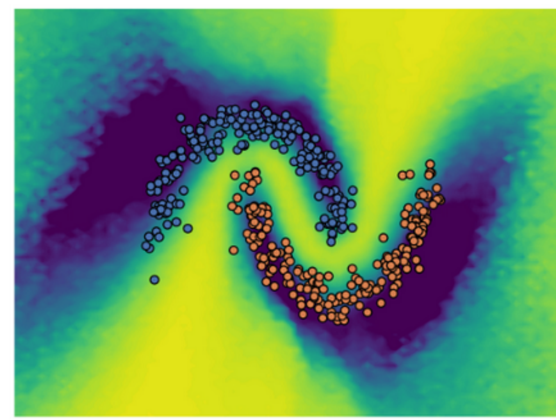

(a) **Ground Truth** - HMC


(b) cSG-HMC


(c) Laplace


(d) MC dropout


(e) Deep Ensembles


(f) **Our Method** - QUAM

Misclassification/OOD detection: different Bayesian DL methods

Table 2: ImageNet-1K results: AUROC using the epistemic uncertainty of a given, pre-selected model (as in Eq. (2)) to distinguish between ID (ImageNet-1K) and OOD samples. Furthermore, we report the AUROC when using the epistemic uncertainty for misclassification detection and the AUC of accuracy over fraction of retained predictions on the ImageNet-1K validation dataset. We also report results for all experiments, using the aleatoric uncertainty of the reference model (Reference).

$\mathcal{D}_{\text{ood}}$ // Task	Reference	cSG-HMC	MCD	DE (LL)	DE (all)	QUAM
ImageNet-O	$.626_{\pm .004}$	$.677_{\pm .005}$	$.680_{\pm .003}$	$.562_{\pm .004}$	$.709_{\pm .005}$	$\mathbf{.753}_{\pm .011}$
ImageNet-A	$.792_{\pm .002}$	$.799_{\pm .001}$	$.827_{\pm .002}$	$.686_{\pm .001}$	$\mathbf{.874}_{\pm .004}$	$\mathbf{.872}_{\pm .003}$
Misclassification	$.867_{\pm .007}$	$.772_{\pm .011}$	$.796_{\pm .014}$	$.657_{\pm .009}$	$.780_{\pm .009}$	$\mathbf{.904}_{\pm .008}$
Selective prediction	$.958_{\pm .003}$	$.931_{\pm .003}$	$.935_{\pm .006}$	$.911_{\pm .004}$	$.950_{\pm .002}$	$\mathbf{.969}_{\pm .002}$

Summary

- We have investigated Bayesian Deep Learning approaches
- Discussed main difference to frequentist approaches
- Can capture uncertainty about parameters
- Usually approximation of parameter posterior
- Variational approach, Monte-Carlo Dropout, Deep Ensembles

Infinite Width Bayesian Deep Networks are Gaussian Processes

- We revisit linear regression:

$$g(\boldsymbol{x}) = \hat{y} = \boldsymbol{w}^T f(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{h}$$

- Now we assume a Gaussian prior over the parameters

$$p(\boldsymbol{w}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{0}, \sigma^2 \boldsymbol{I})$$

- We stack representations and labels of different samples

$$\hat{\boldsymbol{y}} = \boldsymbol{H} w$$

- Then we calculate mean and covariance to define $\boldsymbol{K}$

$$\mathbb{E}[\hat{\boldsymbol{y}}] = \boldsymbol{H}\, \mathbb{E}[w] = \boldsymbol{0}$$

$$\mathrm{Cov}[\hat{\boldsymbol{y}}] = \mathbb{E}[\hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T] = \boldsymbol{H}\, \mathbb{E}[ww^T] \boldsymbol{H}^T = \sigma^2 \boldsymbol{H} \boldsymbol{H}^T = \boldsymbol{K}$$

Gaussian processes

- A Gaussian process is defined as probability distribution over functions $g(\boldsymbol{x})$
- such that the set of values of $g(\boldsymbol{x})$evaluated at a set $\{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^n\}$ is jointly a Gaussian distribution.
- Gaussian distribution is fully determined by its first two moments, **mean** and **covariance**
- The mean vector must be assumed to be zero 0, such that the specification is mostly determined by the **covariance**
- Covariance is determined by the kernel function and thus the pairwise similarity of data points in some space

Connection of neural networks and Gaussian processes

- Discovered by Neal (1994) in his PhD thesis as pessimistic result
- Two-layer network with weights $v_{jd} \sim \mathcal{N}(0, \sigma_w^2)$ and $w_{kj} \sim \mathcal{N}(0, \sigma_w^2)$.
- Pre-activations have identical means and second moments

$$\mathbb{E}[\sum_{d=1}^{D} x_d v_{jd}] = 0 \qquad \mathbb{E}[s_j^2(\boldsymbol{x})] = \mathbb{E}[\sum_{d=1}^{D} x_d^2 v_{jd}^2] = \mathbb{E}[s_l^2(\boldsymbol{x})]$$

- Activations in hidden layer have some mean m_h and variance V_h.
- Distributions of outputs:

$$\mathbb{E}[o_k(\boldsymbol{x})] = \mathbb{E}[\sum_{j=1}^{J} w_{kj} h_j(\boldsymbol{x})] = J\,\mathbb{E}[w_{kj}]\,\mathbb{E}[h_1(\boldsymbol{x})] = 0.$$

$$\mathbb{E}[(o_k(\boldsymbol{x})^2] = \mathbb{E}[\sum_{j=1}^{J} (w_{kj} h_j(\boldsymbol{x}))^2] = J\,\mathbb{E}[w_{kj}^2]\,\mathbb{E}[(h_1(\boldsymbol{x}))^2] = J\sigma_w^2 V_h.$$

Connection of neural networks and Gaussian processes

- We can use the *Central Limit Theorem* to show that outputs are Gaussian distributed (similar to our derivation of SNNs)
- The joint distribution of two data points then is

$$\mathbb{E}[o_k(\boldsymbol{x}_n) o_k(\boldsymbol{x}_m)] = \sum_{j=1}^{J} \mathbb{E}[w_{jk} h_j(\boldsymbol{x}_n) w_{jk} h_j(\boldsymbol{x}_m)] = \sigma_w^2 V_h K(\boldsymbol{x}_n, \boldsymbol{x}_m)$$

- Gaussian for multiple data points, thus a Gaussian process whose covariance depends on kernel between data points
- Covariance between two different output units is zero
- Connection is often used for Deep Learning theory
  - Also for *Neural Tangent Kernels*